

A DISTRIBUTED SIMULATION SANDBOX FOR DECENTRALIZED CONTROL ALGORITHMS IN MARINE DRONE SWARMS

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Masters Project in Engineering (Drones and Autonomous Systems)
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Abstract

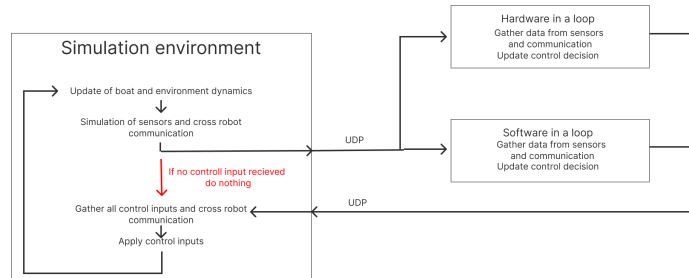
This thesis proposes the design and implementation of a high-performance simulation sandbox for developing and testing decentralized control algorithms for autonomous sailing drone swarms. The core of the project is a simulation environment. The simulator models key marine dynamics, including wind and sea conditions, and incorporates virtual sensors, along with inter-drone communication channels.

The motivation for this work is its application in maritime search and rescue (SAR). By leveraging a swarm of inexpensive, autonomous drones, the time to locate a drowning victim can be significantly reduced, dramatically increasing the chances of survival.

As of October 2025, there are no known existing solutions that combine simulation, SITL/HITL capabilities, and decentralized control for marine drone swarms, making this project a pioneering effort in autonomous maritime robotics. Existing tools tend to target aerial drones or are general-purpose robotics simulators and require heavy modification to support marine dynamics and large-scale swarm experiments. A related project is YARA[?], a Gazebo-based sailing drone simulator, but it lacks SITL/HITL support and is not optimized for extensive swarm simulations.

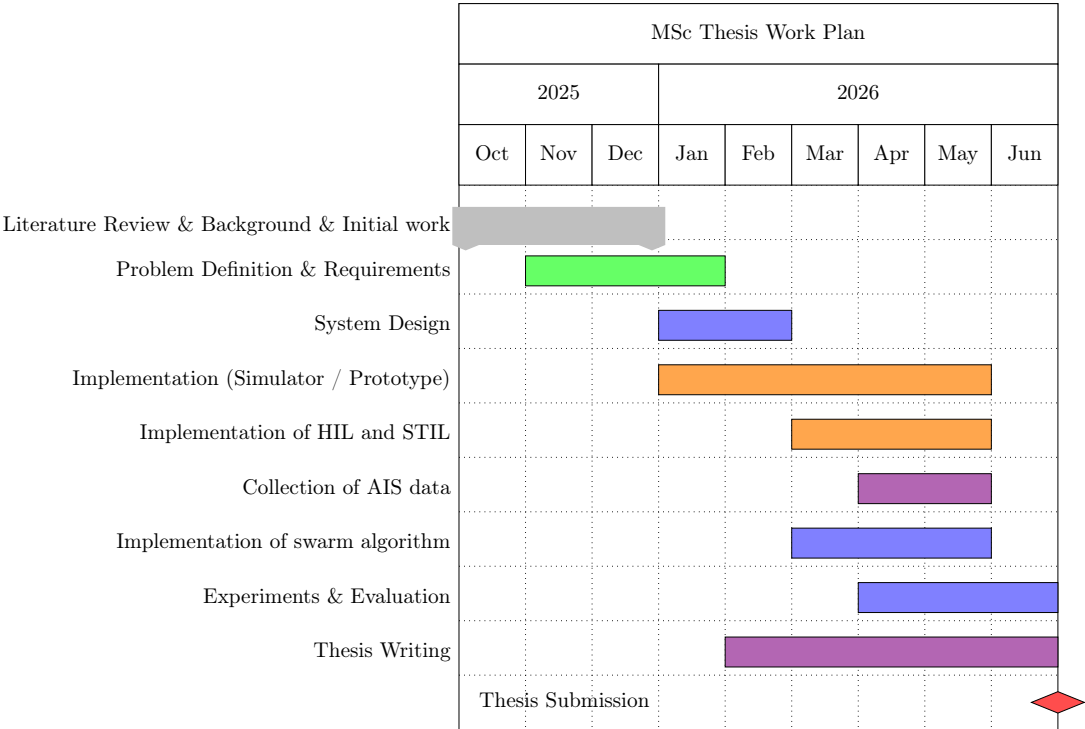
This work will implement a cross-robot communication channel model with signal attenuation, delays, and packet dropouts. The simulator will provide sensor models for IMU, GPS, and AIS (Automatic Identification System) and will support replaying historical AIS datasets to populate and simulate real vessel traffic within the environment. The platform will also allow isolating and instantiating independent swarms to enable large-scale training and evaluation of genetic algorithms and neural-network based controllers.

The simulator will be validated through a series of experiments, including benchmarking decentralized control algorithms for formation keeping, collision avoidance, and cooperative search tasks. The performance of the simulator will be assessed based on its computational efficiency, scalability, and fidelity in replicating real-world marine conditions.



Proposed simulation dataflow.

WORK SCHEDULE / GANTT CHART



0.1 Drone Ship Physics Model

Very important part of my project is to have a good physics model of the drone ship. That model will be used to simulate IMU and GPS sensor readings, as well as to provide a realistic environment for testing control algorithms.

First step is to model the buoyancy of the ship. The ship will be modeled as a rigid body with a certain mass, width, length, and draft. Currently model does not assume any particular shape for the ship, but it will be important to consider the shape in future iterations to improve the accuracy of the model. The buoyancy force will be calculated based on the 4 points of the ship: the bow, stern, port, and starboard. The buoyancy force will be calculated as the weight of the displaced water, which is equal to the volume of the submerged part of the ship multiplied by the density of water and gravity.

Same points will be used to calculate the wave drag force. In each of 4 points there is a wave drag force acting in the normal direction to the surface of the water. Forces are scaled to the size of the ship.

This approach captures the essential physics of the problem while keeping the model computationally efficient. Placement of these points is visible in the figure below 0.1.

That model will be used to simulate the motion of the ship in response to control inputs and environmental forces.

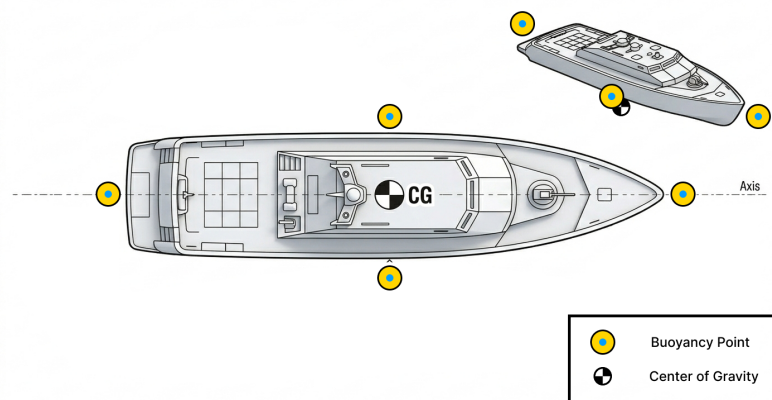


Figure 0.1: Ship buoyancy model diagram.